



Do AR museum tours on visitors' phones improve wayfinding,
engagement, and learning versus a printed label tour?

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Executive Summary

This feasibility study evaluates the practicality of an individual research component that investigates whether a QR code based mobile augmented reality (AR) fossil tour, enhanced with a Virtual AR Guide, can improve visitor navigation, engagement, and learning in museum environments when compared to traditional printed-label interpretation methods.

The developed mobile AR application consists of two primary features. The first is a Normal AR mode, implemented as an open, gallery style experience that allows users particularly children to place and interact with animated 3D animal models without scanning QR codes. The second and core research feature is the AR Fossil Tour, a location-restricted experience designed specifically for museum environments. This feature is activated by scanning a QR code at the museum entrance, confirming the user's physical presence inside the museum and enabling guided AR based exploration.

In the AR Fossil Tour, users are first introduced to a Virtual AR Guide who provides a brief welcome and explains how assistance can be accessed during the tour. Users can then select the prehistoric animals they wish to explore and navigate between corresponding fossil exhibits using AR floor based directional arrows. Instead of relying on automatic audio narration, all educational information is delivered through direct interaction with the Virtual AR Guide, allowing users to request explanations about animal characteristics, behavior, habitat, and historical context at their own pace. Voice-based interaction is supported using free and open-source technologies to ensure technical feasibility, ethical safety, and suitability for child users.

At the time of this feasibility study, the Normal AR mode, QR code scanning functionality, and core AR infrastructure have been successfully implemented and tested on mid-range Android devices (Android 7 and above). The AR Fossil Tour navigation logic has been implemented, while the Virtual AR Guide interaction and voice-based communication features are under development. Planned Bluetooth Low Energy (BLE) beacon integration remains a future enhancement for indoor positioning support.

This research component emphasizes a user-controlled learning model that aligns with real-world museum behavior, where visitors choose when and how to engage with explanatory content. By allowing users to actively request information through the Virtual AR Guide, the system aims to reduce cognitive overload, support inquiry driven exploration, and improve overall visitor experience while maintaining a low cost, scalable solution suitable for Sri Lankan Museum contexts.

The feasibility analysis evaluates the individual research component across technical, market, financial, operational, and ethical dimensions. Based on the current implementation and system design, the study concludes that the proposed AR Fossil Tour with a Virtual AR Guide is technically feasible, financially viable, operationally practical, and ethically sound, and is therefore suitable for full implementation and empirical evaluation.

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1. Introduction

1.1 Background

Museums play a vital role as informal learning environments by preserving and presenting scientific, historical, and cultural knowledge to the public. However, many museums in Sri Lanka continue to rely primarily on printed labels and static display panels to communicate information about exhibits. While these methods are cost-effective and easy to maintain, they often provide limited support for visitor engagement, intuitive navigation, and sustained learning, particularly for children and younger audiences who are increasingly accustomed to interactive digital experiences.

Recent developments in mobile augmented reality (AR) have demonstrated significant potential for enhancing museum experiences by overlaying digital information directly onto physical spaces using visitors' personal smartphones. Mobile AR solutions reduce the need for expensive hardware installations and permanent infrastructure changes, making them especially suitable for resourceconstrained museum environments. QR code-activated AR experiences offer an additional advantage by providing a simple and scalable method to trigger location-specific content without requiring complex positioning systems.

Despite these advantages, many existing AR museum applications rely on automatic audio narration or linear content delivery models. Such approaches can reduce user autonomy, overwhelm visitors with information, and limit opportunities for inquiry-based learning. Visitors often differ in their interests, attention spans, and learning preferences, highlighting the need for systems that allow users to control both the timing and depth of information presented.

To address these limitations, this research component explores a QR code-based mobile AR Fossil Tour that incorporates a Virtual AR Guide to mediate information delivery. Rather than presenting information automatically, the Virtual AR Guide provides explanations only when requested by the user, closely resembling the behavior of a real museum guide. This design supports self-paced exploration, encourages curiosity-driven interaction, and aims to improve overall visitor engagement while remaining technically and financially feasible for Sri Lankan Museum contexts.

1.2 Objective of the Feasibility Study

The objective of this feasibility study is to evaluate whether a QR code based mobile augmented reality (AR) Fossil Tour, enhanced with a Virtual AR Guide, can be realistically designed, developed, and deployed as an individual research component within a museum environment.

Specifically, this study aims to determine whether the proposed system can be implemented using mobile AR technologies while maintaining stable performance on mid range smartphones commonly used by museum visitors. The feasibility study also seeks to assess whether replacing automatic audio narration with a user-controlled Virtual AR Guide can improve usability and support inquiry-based learning without introducing excessive technical complexity.

In addition, the study evaluates the practicality of integrating AR navigation, animal selection based tour customization, and voice based interaction using free and open-source tools. The objective is to ensure that the system remains financially viable, operationally practical for museum deployment, and ethically appropriate, particularly for child users. The findings of this feasibility study will inform decisions regarding full system implementation and subsequent empirical evaluation of learning and engagement outcomes.

1.3 Scope of the Study

The scope of this feasibility study defines the specific boundaries within which the proposed QR code based mobile augmented reality (AR) Fossil Tour with a Virtual AR Guide is evaluated. The study focuses on determining whether the system can be successfully developed, deployed, and evaluated as an individual research component within a museum environment using mobile AR technologies.

This feasibility study examines the proposed solution across key feasibility dimensions, including technical, market, financial, operational, and ethical considerations. Technical feasibility focuses on the use of Unity, AR Foundation, QR code scanning, AR navigation, and voice-based interaction using free and open-source tools on mid-range mobile devices. Market feasibility considers the suitability of the system for children, students, teachers, and general museum visitors, with emphasis on usability, engagement, and acceptance of guided AR interaction as an alternative to printed labels.

Financial feasibility within this scope evaluates whether the system can be implemented without specialized hardware and with minimal operational costs, relying primarily on QR codes and optional Bluetooth Low Energy (BLE) beacons. Operational feasibility considers deployment within an existing museum setting without structural modifications, self-guided visitor usage, and limited staff involvement. Ethical and accessibility feasibility addresses user privacy, responsible microphone usage, child-friendly interaction design, and inclusive access to information.

This feasibility study does not assess final learning outcomes or conduct comparative analysis against printed label tours, as these aspects are intended to be evaluated during the post-implementation empirical study. The scope is limited to assessing whether the proposed AR Fossil

Tour with a Virtual AR Guide is feasible for full development and controlled evaluation within an academic research context.

2. Problem Statement

Traditional museum tours that rely on printed labels and static information panels often fail to provide intuitive wayfinding, sustained engagement, or adaptive learning experiences for visitors. In many cases, visitors struggle to navigate exhibition spaces efficiently, overlook important exhibits, or disengage due to text-heavy explanations that do not accommodate different learning preferences, attention spans, or levels of prior knowledge.

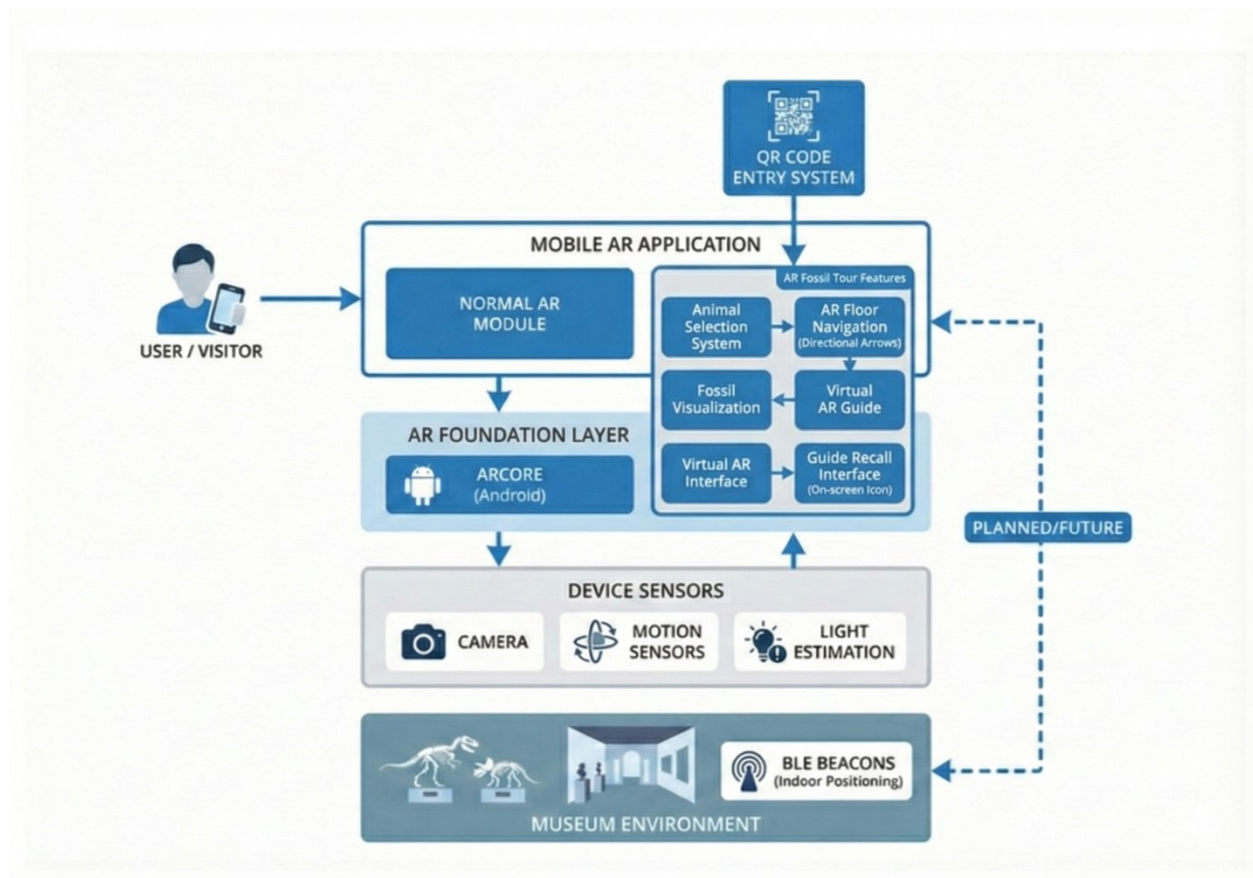
Although digital and interactive museum systems exist, many of these solutions depend on costly hardware installations, permanent infrastructure changes, or continuous staff involvement. Such requirements make them impractical for widespread adoption in Sri Lankan Museum contexts, where budgetary constraints and operational limitations are significant considerations.

Furthermore, existing mobile augmented reality (AR) museum applications frequently rely on automatic audio narration or linear content delivery. While informative, these approaches can reduce visitor autonomy, interrupt natural exploration, and increase cognitive load, particularly for child users. There is a clear need for a mobile-first, low-cost solution that supports self-guided navigation and allows visitors to actively request information when needed.

This research addresses the identified problem by evaluating the feasibility of a QR code based mobile AR Fossil Tour enhanced with a Virtual AR Guide. The proposed approach aims to improve navigation, engagement, and inquiry-based learning while remaining technically, financially, and operationally viable for real museum environments.

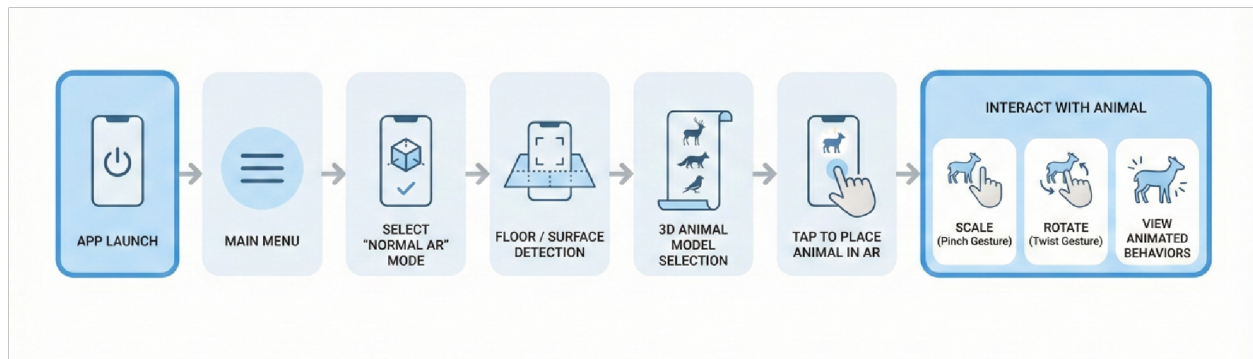
3. Project Description

The proposed solution is a mobile augmented reality (AR) application developed using Unity and AR Foundation, optimized for use on mid-range Android smartphones. The application is designed to support informal learning in museum environments by enhancing traditional fossil exhibits with interactive AR-based guidance and information delivery.



Appendix A: High Level System Architecture of the AR Tour Application

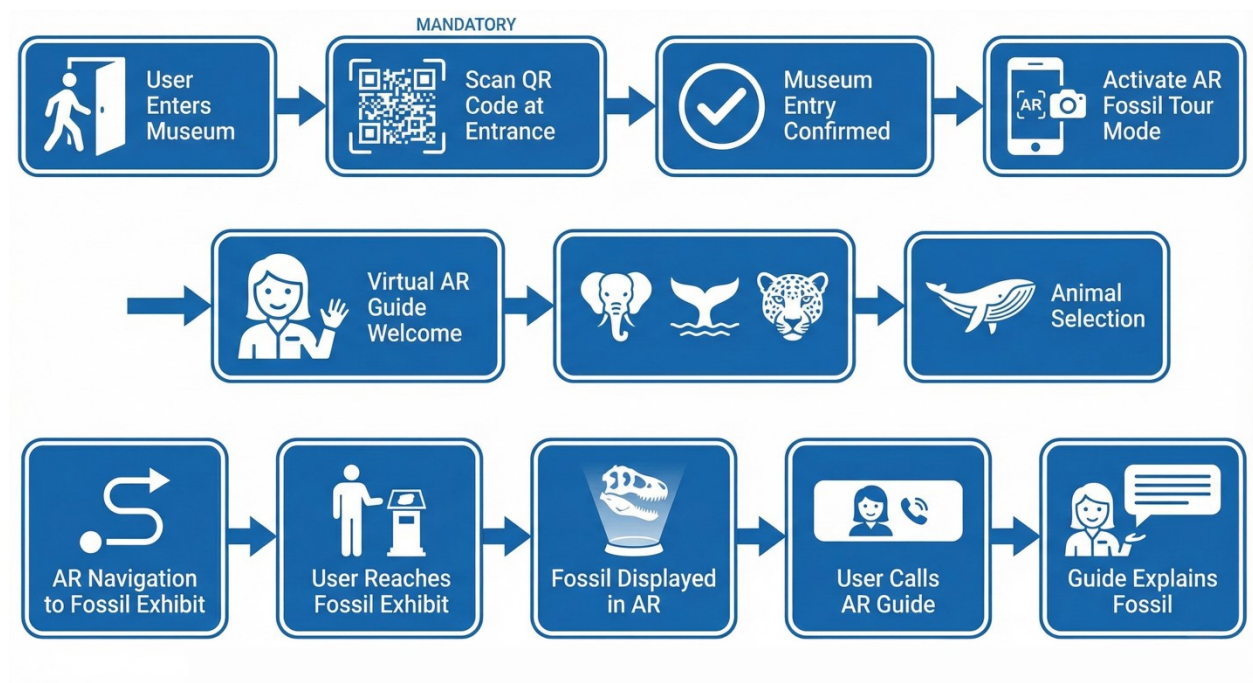
The application consists of a shared AR infrastructure that supports two primary features: a Normal AR mode and an AR Fossil Tour. Both features utilize the same underlying AR pipeline for camera access, plane detection, 3D rendering, and user interaction, while serving different usage contexts. The Normal AR mode functions as an open, gallery style experience that allows users to place and interact with animated 3D animal models without location restrictions. This feature validates core AR interaction techniques and supports engagement outside museum environments.



Appendix B: Normal AR User Interaction Flow

The AR Fossil Tour is the core research component and is designed as a location-restricted experience intended specifically for museum use. The tour is activated by scanning a QR code at the museum entrance, which confirms the user's physical presence within the exhibition space. Once activated, users are guided through the museum using AR floor based directional arrows that lead to selected fossil exhibits.

A key feature of the AR Fossil Tour is the integration of a Virtual AR Guide that mediates all educational interaction. Instead of relying on automatic audio narration, the system allows users to summon the guide at any time to request information. Users can select specific animals they wish to explore and interact with the guide through voice-based queries to learn about animal characteristics, behavior, habitat, and historical context. This guide-centric interaction model supports self-paced, inquiry-driven learning and closely resembles real-world museum guide behavior.



Appendix C: AR Fossil Tour Activation and Navigation Flow

The project is designed to operate without specialized hardware or permanent installations, relying primarily on QR codes and mobile device capabilities. Optional Bluetooth Low Energy (BLE) beacon integration is considered as a future enhancement to support more precise indoor positioning. The overall system emphasizes low cost, scalability, and ethical use, making it suitable for deployment and evaluation in Sri Lankan Museum contexts as part of an academic research study.

4. Feasibility Analysis

This section evaluates the feasibility of the proposed AR Fossil Tour with a Virtual AR Guide across five key dimensions: technical, market, financial, operational, and ethical and accessibility feasibility. The analysis is based on the current implementation status, system design, and planned development approach of the individual research component.

4.1 Technical Feasibility

The proposed system is developed using Unity and AR Foundation, leveraging ARCore for mobile augmented reality functionality on Android devices. Core AR features such as plane detection, camera tracking, 3D model rendering, and gesture-based interaction have already been successfully implemented and validated through the completed Normal AR mode. This provides strong evidence that the underlying AR infrastructure is stable and capable of supporting the more complex AR Fossil Tour functionality.

The AR Fossil Tour relies on QR code scanning to activate the experience and confirm user presence within the museum. QR scanning functionality has already been implemented and tested, demonstrating reliable performance under typical indoor lighting conditions. AR navigation is implemented using predefined waypoints and floor-based directional arrows, avoiding the need for complex real-time pathfinding. Planned Bluetooth Low Energy (BLE) beacon integration is supported by existing Android APIs and can be incorporated without major architectural changes.

The Virtual AR Guide and voice based interaction system are technically feasible using free and open-source tools. Speech to text processing is supported using Whisper, intent recognition is handled through rule based logic or free tier natural language processing tools, and text to speech output is generated using Coqui TTS. These technologies are well documented, compatible with Unity through local or REST based integration, and do not introduce dependency on paid cloud services.

Conclusion: The system is technically feasible within the constraints of a university level research project and is suitable for full implementation and testing.

4.2 Market Feasibility

The target users for the proposed system include school children, students, teachers, families, and general museum visitors. These user groups increasingly expect interactive and technology assisted learning experiences, particularly when engaging with scientific and historical content. Mobile AR is well aligned with these expectations, as it leverages devices that visitors already own and are familiar with.

The guide-centric interaction model is especially suitable for children and first-time museum visitors, as it allows users to request information at their own pace rather than being subjected to continuous or unwanted narration. The ability to select specific animals of interest and receive guided explanations supports personalized exploration and increases perceived usefulness.

Conclusion: The proposed AR Fossil Tour demonstrates strong user relevance and acceptance potential within educational and museum contexts.

4.3 Financial Feasibility

The financial requirements of the proposed system are minimal. The application is developed using free or academic versions of Unity and open-source software libraries. No specialized hardware installations are required, as the system operates on visitors' personal smartphones.

Physical deployment costs are limited to the printing and placement of QR codes and, if implemented, a small number of Bluetooth Low Energy beacons. These costs are significantly lower than those associated with traditional guided tour systems or permanent digital installations. Maintenance costs are also minimal, as content updates can be managed digitally without physical modifications.

Conclusion: The project is financially viable and well suited for implementation within typical academic and museum budget constraints.

4.4 Operational Feasibility

The proposed system is designed to integrate smoothly into existing museum operations. QR codes can be placed near entrances and exhibits without altering the physical structure of the museum. Visitors are able to self navigate the tour using their own devices, reducing the need for continuous staff involvement or supervision.

The Virtual AR Guide further supports operational feasibility by providing explanations and guidance that would otherwise require human guides. The system can be deployed incrementally, starting with a limited number of exhibits and expanding as needed. The offline first design also ensures stable operation in environments with limited or unreliable internet connectivity.

Conclusion: The system is operationally practical and suitable for real museum deployment with minimal disruption.

4.5 Ethical and Environmental Feasibility

Ethical considerations are carefully addressed in the system design. The application does not collect or store personal user data. Microphone access is used only for active voice interaction and requires explicit user permission. All voice processing is designed to be session based, ensuring user privacy.

Accessibility considerations include a child-friendly user interface, large touch targets, simple navigation cues, and the option to interact with the system without voice input if required. The guide-based interaction model allows users to control the amount of information they receive, reducing cognitive overload and supporting inclusive learning experiences.

Conclusion: The proposed system is ethically sound and accessible for a wide range of users.

5. Risk Analysis

The development and deployment of the proposed AR Fossil Tour with a Virtual AR Guide present several potential risks that may affect system performance, usability, and evaluation outcomes. Identifying these risks at the feasibility stage allows appropriate mitigation strategies to be considered during implementation.

One potential risk is instability in AR tracking within indoor museum environments. Factors such as inconsistent lighting, reflective surfaces, or limited visual features may affect plane detection and tracking accuracy. This risk is mitigated by using optimized lighting conditions, carefully selecting tracking areas, and implementing predefined waypoints rather than relying solely on continuous spatial mapping.

QR code scanning reliability also presents a moderate risk, particularly if codes are poorly placed or exposed to inadequate lighting. To reduce this risk, high contrast QR codes can be used and positioned at accessible heights near entrances and exhibits. Clear visual instructions can further support successful scanning.

Voice recognition inaccuracies represent another potential challenge, especially when users speak softly, have different accents, or when background noise is present. This risk is mitigated by using intent-based interaction rather than open-ended conversation, allowing users to repeat queries, and providing alternative interaction methods if voice input fails.

There is a potential risk of increased cognitive load for younger users if too much information is delivered at once. This is addressed by designing short, simple guide responses and ensuring that all interactions with

the Virtual AR Guide are optional and user-controlled. This approach allows users to engage at their own pace.

Device performance limitations on older or low-end smartphones may impact rendering performance or audio playback. This risk is mitigated by optimizing 3D models, reducing polygon counts, managing audio resources efficiently, and testing the system on mid-range devices to ensure acceptable performance.

User confusion during navigation is another possible risk, particularly for first-time AR users. This can be reduced through clear onboarding instructions, consistent AR arrow guidance, and easy access to the Virtual AR Guide for assistance at any point during the tour.

Finally, if Bluetooth Low Energy (BLE) beacons are implemented for indoor positioning, signal inconsistency may affect navigation accuracy. This risk can be mitigated through proper calibration, limited reliance on BLE data, and maintaining a hybrid approach that combines QR-based activation with waypoint-based AR navigation.

Overall, while several risks have been identified, they are considered manageable within the scope of an academic research project. With appropriate design choices and mitigation strategies, these risks are unlikely to significantly hinder successful implementation or evaluation of the proposed system.

6. Recommendations

Based on the findings of this feasibility study, several recommendations are proposed to support the successful implementation and evaluation of the AR Fossil Tour with a Virtual AR Guide as an individual research component.

It is recommended that development proceed in a phased manner, prioritizing the completion of the Virtual AR Guide interaction system and the integration of voice-based communication. Implementing the guide using a controlled, intent-based interaction model should be prioritized to ensure accuracy, usability, and suitability for child users.

Further refinement of the AR navigation system is recommended to ensure clear and intuitive wayfinding within the museum environment. Floor-based directional arrows should be tested in different lighting conditions and exhibition layouts to ensure consistent visibility and effectiveness.

User interface and interaction design should be evaluated with particular attention to children and first-time AR users. Clear onboarding instructions, simple guide dialogue, and easily accessible help mechanisms are recommended to reduce cognitive load and improve overall user experience.

It is also recommended that usability testing be conducted with a small group of representative users, such as students or museum visitors, prior to full deployment. Feedback from these sessions can be used to refine guide behavior, voice interaction clarity, and navigation flow.

If indoor positioning accuracy becomes a limitation, Bluetooth Low Energy (BLE) beacon integration may be explored as an enhancement. However, this should be implemented cautiously and evaluated alongside existing waypoint-based navigation to avoid unnecessary complexity.

Finally, it is recommended that detailed documentation of system design decisions, user interactions, and technical performance be maintained throughout development. This will support subsequent empirical evaluation and strengthen the overall research contribution of the project.

7. Conclusion

This feasibility study evaluated the practicality of implementing a QR code-based mobile augmented reality Fossil Tour enhanced with a Virtual AR Guide as an individual research component. The analysis considered the system across technical, market, financial, operational, and ethical and accessibility dimensions to determine whether it can be successfully developed and deployed within a museum environment.

The findings indicate that the proposed system is technically feasible using current mobile AR technologies, with the completed Normal AR mode providing evidence of stable AR tracking, interaction handling, and performance on mid-range devices. The redesigned guide-centric interaction model offers a realistic and user-controlled alternative to automatic narration, supporting inquiry-driven learning without introducing excessive technical complexity.

From a market and operational perspective, the system aligns well with the needs of museum visitors, particularly children and educational groups, by enabling self-guided exploration using personal mobile devices. Financial analysis confirms that the solution is cost-effective, requiring no specialized hardware and relying primarily on QR codes and optional enhancements.

Ethical and accessibility considerations have been appropriately addressed through responsible microphone usage, avoidance of personal data collection, and child-friendly interaction design. Identified risks are manageable and can be mitigated through careful implementation and testing.

In conclusion, the proposed AR Fossil Tour with a Virtual AR Guide is feasible for full implementation and controlled evaluation within an academic research context. The system provides a strong foundation for subsequent empirical studies aimed at assessing its impact on visitor engagement, navigation efficiency, and learning outcomes in museum environments.

8. Appendices

The appendices provide supporting diagrams and descriptions that supplement the main content of this feasibility study. They include the high-level system architecture and detailed user interaction flows for both the Normal AR mode and the AR Fossil Tour, offering additional clarity on system design and operation without disrupting the main discussion.

Appendix A: High-Level System Architecture

This appendix presents the high-level system architecture of the proposed mobile augmented reality application. The architecture illustrates the overall structure of the system and how its main components interact to support both the Normal AR mode and the AR Fossil Tour. The diagram highlights the shared AR infrastructure built on the AR Foundation layer, the use of device sensors for AR tracking, and the role of QR code activation in enabling the AR Fossil Tour within the museum environment.

Appendix B: Normal AR User Interaction Flow

This appendix describes the user interaction flow for the Normal AR mode of the mobile augmented reality application. The interaction begins when users select the Normal AR mode, after which the system performs floor detection to identify a suitable surface. Users can then select and place a 3D animal model in their physical environment and interact with it using gesture-based controls such as scaling and rotation. The Normal AR mode also allows users to capture and save images of the AR model, demonstrating the core AR interaction capabilities of the application.

Appendix C: AR Fossil Tour User Interaction Flow

This appendix outlines the user interaction flow of the AR Fossil Tour, which is designed specifically for museum environments. The tour is activated by scanning a QR code at the museum entrance, confirming the user's presence within the exhibition space. Users are introduced to a Virtual AR Guide and can select the prehistoric animals they wish to explore. The system then guides users to selected fossil exhibits using AR floor-based directional arrows. At each exhibit, users can summon the Virtual AR Guide to request information about the fossil, supporting a user-controlled, inquiry-driven learning experience.